

Supporting information

S2 Algorithm optimisation

5 A pairwise comparison of trait values costs more computation time than a single comparison to the mean. As expected, the simulations for the pairwise competition model were slower (around 4 times) than for the mean competition model. Yet, the computation time is a critical issue in a context of large numbers of simulation experiments required for classifier training, and this necessitated code optimization.

Model	Before optimization: [29] with pairwise competition added by this study	After optimization: this study	Ratio
Neutral	9 ± 2	24 ± 3	0.375
Mean competition	456 ± 114	89 ± 11	5.14
Pairwise competition	1944 ± 258	323 ± 40	6.02
Environmental filtering	80 ± 18	70 ± 10	1.14

S2 Table 2: Comparison of the speed of the simulations for different version

□ **of the code** (mean value per run, $\pm$ standard deviation, in seconds). Results are for 50 simulations of 200 generations on a single core.